Power Budget

Team Number: 305

Project Name: HMI Subsystem

Team Member: Christo

Version: 1

All Major Components	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	ESP32-S3-WROOM	ESP32-S3-WROOM	+3.0 - 3.6V	1	500	500	mA
	0.96in OLED LCD	CJJ_LCD 0.96	+3.3V	1	20	20	mA
	Status LEDs x4	Generic LED	+3.3V	4	7	28	mA
	LM2596S-3.3	LM2596S-3.3	+4.5 - 40V	1	3000	3000	mA

+3.3V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	ESP32-S3-WROOM	ESP32-S3-WROOM	+3.0 - 3.6V	1	500	500	mA
	0.96in OLED LCD	CJJ_LCD 0.96	+3.3V	1	20	20	mA
	Status LEDs x4	Generic LED	+3.3V	4	7	28	mA
					0	0	
					Subtotal	**548**	mA
					Safety Margin	**25%**	
					Total Current Required on +3.3V Rail	**685**	mA

Regulator	LM2596S-3.3	LM2596S-3.3	+4.5 - 40V	1	3000	3000	mA
					Total Remaining Current Available on +3.3V Rail	**2315**	mA

External Power Source	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source Selectior	9V 3A Wall Adapter	(full part number)	9V DC	+9V	3000	3000	mA
Power Rails Connected to External Power Source	LM2596S-3.3	LM2596S-3.3	+4.5 - 40V	+3.3V	3000	3000	mA
					Total Remaining Current Available on External Power Source	**2315**	mA

Notes:

LM2596S-3.3 is rated for 3A continuous output. 25% safety margin applied to all rail calculations. ESP32-S3 peak draw during WiFi transmission used as max current figure.